

CHEM&121: Introduction to Chemistry

Why This Course, Why You

Chemistry is the foundation of the human body itself — the foods you eat and the environment around you determines physiological impacts and outcomes at a (bio)chemical level. This course exists to build the specific chemistry reasoning health professionals actually use like dosage calculations, preparing solutions, gases, and data and information literacy – things that shows up constantly in clinical work. So, this isn't chemistry for its own sake — it's a foundation your future career gets built on.

Chemistry also puts a spotlight on how comfortable you are with learning itself. This isn't easy for anyone, because it's more than recalling facts and getting things right — it's being comfortable in the space between what you know and what you don't yet. Real understanding comes with time and repetition, and there are no shortcuts. Everything in this syllabus is built around getting you through that process, not around catching you out.

Start Here

If you read one part of this syllabus, read this one.

Item	Details
What it costs	Chemfolio (\$19.50) + CHEM&121 Lab Manual (\$16.70) (Clark Bookstore) + ALEKS (paid through your course registration).
When things are due	Wednesday through Sunday each carry something — see Weekly Flow below for the exact rhythm. Once you learn it, it repeats every module.
Time to budget	About 10–12 hours a week — most of it on Canvas, plus your 4-hour on-campus lab.
How to reach me	Canvas Inbox only — I don't use email during the term. I reply within 24 hours on weekdays.
The two big ones	Weekly lab attendance (required every week, no exceptions) and the Final TYC (Teach Your Community) Project — 10% of your grade, in place of a final exam.
The AI rule	No AI-generated content in anything you submit — ever, no exceptions. See the AI Agreement below for the full policy.
If you fall behind	No explanation needed. Late work loses 1% per hour after the due date, down to zero after about 5 days — then it can still be submitted for zero credit. See Late Work below for the full policy.

Your grade in one line: 3-2-1 Team Posts 15% · OOL Discussions 15% · Lab Experiments 25% · Quizzes 20% · ALEKS Problem Sets 15% · Final TYC Project 10%.

Course Terms You'll See

A few abbreviations come up constantly in this course. Here's what they mean, so you're not guessing the first time you see them:

Term	What it means
3-2-1	Your weekly Team Discussion format — named for its posting structure (three problem areas, two areas of interest, one insight from the video key).
OOL	Outside Of Lecture — a separate weekly class discussion, not the 3-2-1.
TYC	Teach Your Community — your Final Project, replacing a final exam.
ALEKS	Your online adaptive homework and textbook-access platform.
DAC	Disability Access Center — Clark's office for accommodations.
Chemfolio	Your required lecture/discussion workbook — you buy this!
TL;DR	"Too long; didn't read" — a quick plain-language summary. Wherever you see TL;DR, that line tells you the gist before the official/detailed text below it.

Your First Week

1. Attend your scheduled lab session. Missing it means being dropped from the course.
2. Log into Canvas and read the announcements at the top of the course.
3. Work through Getting Started: Course Orientation, including the ALEKS/Office/OneDrive setup guide — both need to be done by the end of your first lab, so budget the time.
4. Read this syllabus, especially the Start Here box and the Course Agreements section.
5. Confirm you have both your Chemfolio and lab manual from the Clark Bookstore before class — your first 3-2-1 team post is due that Wednesday.

Course Information

Item	Details
Course Title	CHEM&121: Introduction to Chemistry (Pre-Health)
Course Nbr/Sec	18405/18406, 18407/18408
Quarter/Year	Fall 2026 (September 21 – December 10, 2026)
Location/Time	Lab: STEM Building 222, Mo/We 2:00–3:50 PM (18405) or Mo/We 5:00–6:50 PM (18407). Lecture/discussion: online, asynchronous.
Course Site	Canvas Learning Management System
Course Modality	Hybrid — asynchronous online lecture/discussion + biweekly on-campus lab
Credit Hours	5 credits (3.0 hours lecture-equivalent / 4.0 hours lab)
Prerequisites	MATH 96 (grade of "C" or higher), or placement into Math level 45.
Course Description	What Clark's Course Catalog says: Topics in general chemistry applicable to students seeking a 2-year degree in the health-occupations fields. Unit-factor method is applied to problem solving. Topics covered include units of measurement, atomic structure, chemical bonding, energy, the mole concept, nomenclature of inorganic compounds, writing and balancing equations, properties of gases, solutions and colloids, reaction rates and equilibrium, acids, bases and salts, radiation and health.

Item	Details
	What that means in practice: You'll learn to think like a health professional needs to think about chemistry — using math and unit conversions to solve real clinical-style problems, not just memorizing facts. Expect this course to feel like a math class and a foreign-language class rolled together: we use conversions from day one, every week, in both discussion and lab.

Welcome to Class!

TL;DR: No lectures — everything happens through weekly team discussions and hands-on labs. Show up, stay engaged, and ask for help before you're stuck.

Welcome, welcome, welcome! I'm Dr. B (Dr. Karl Bailey, he/him), and I'm so glad to be your Chemistry instructor this term. However you landed in this course, I'm glad you're here — let's get into how it actually runs, week to week.

Dr. B's Note: *Most students who stick with the weekly rhythm and stay engaged do well. The difference isn't talent, it's showing up consistently. Give yourself a fighting chance.*

What preparation/skills do I need for this class?

CHEM is a math class combined with a foreign language course, and we move fast. We use math conversions from day one and keep using them every week, in both discussion and lab. If your math skills need more than a light dusting-off, consider brushing up before or alongside this course — most students taking CHEM&121 find the math the hardest part, not the chemistry itself.

What can I expect in class?

This course is entirely discussion-based online — there are no traditional lectures. Instead, you'll work through weekly Activities from your Chemfolio in a 3-2-1 Team Discussion format and apply what you're learning hands-on in your required weekly on-campus labs. Labs involve real data collection and analysis, not cookbook steps — more on exactly how in Weekly Flow and the Grading section below.

Dr. B's Note: *If you've taken a science course before, don't assume this one works the same way: there are no traditional lectures, quizzes happen in person during lab, and labs are real science, not a recipe to follow. More on exactly how below.*

What if I need an accommodation?

Clark College and the Disability Access Center (DAC) office assist students with disabilities in pursuing their educational goals. DAC can provide classroom accommodations, as well as serve as a resource to the campus community in striving to make Clark College both an accessible and hospitable place for students with disabilities. Interested in getting accommodations? Begin with [viewing the steps for requesting accommodations](#).

Weekly Flow

TL;DR: Something is due nearly every day, Wednesday through Sunday — same pattern every week once you learn it.

Canvas Modules are your home base. This rhythm repeats every module — once you know this table, you know the whole term.

Day	What's due
Monday	New module opens
Wednesday	3-2-1 Team post + upload due; OOL discussion post due; Lab Results (20 pts) due
Thursday	3-2-1 Team Lead Summary post due (Team Lead only, that week)
Friday	OOL replies due; Lab Findings (10 pts) and Conclusions (20 pts) due
Sunday	3-2-1 Team post + replies to all 3 teammates + upload due; ALEKS homework due; Lab Background (10 pts) and Methods (40 pts) due

Attend lab both days assigned in your schedule.

Quizzes are the one exception to this weekly pattern — they're in-person during lab, in Weeks 2, 5, 7, 9, and 10.

Dr. B's Note: *Prioritize content learning in your Chemfolio with your team first. Labs are based on the assigned content from the module.*

About Your Instructor

TL;DR: Canvas Inbox only, no email — I reply within 24 hours on weekdays.

Item	Details
Instructor Name and pronouns	Dr. Karl Bailey, Ph.D. (he/him)
How I like to be addressed	Dr. B
Communication	Canvas Inbox only — I do not use email during the term. Not a current student? Reach me at kbailey@clark.edu or 360-992-2313.
Help Hours	Monday & Wednesday, 1:00–2:00 PM and 4:00–5:00 PM (1 hour before each lab section)
Preferred Mode of Communication	Canvas Inbox — I reply within 24 hours on weekdays.

What can you expect from me?

I've spent more than 25 years building and refining active-learning chemistry courses putting students and their learning at the center of my teaching whether in person or online. I've designed everything in this course with you in mind based on what my students tell me works. The first big part of that is learning to struggle when learning anything new. It's normal. So stick with it and you'll find your footing and confidence with it. Reach out for help directly before things get away from you. I respond to Canvas messages within 24 hours on weekdays.

"Our job as educators is not to prepare you for the real world. You already live in the real world. Our job is to empower you to change it." — Dr. B

What Course Materials Will I Need?

TL;DR: Chemfolio + your lab manual (Clark Bookstore) + ALEKS (set up automatically in Canvas).

Item	Details
Texts/Other learning materials	Chemfolio (lecture workbook) — required, drives our 3-2-1 discussion format. (\$19.50, edition/ISBN TBD) CHEM&121 Lab Manual (lab manual) — required. (\$16.70, edition/ISBN TBD) Composition Notebook, college ruled, 7.5 x 10 in — required. Scientific or Graphing Calculator — required physical calculator, no alternative electronic devices. If you'll be taking Math 146 Statistics, we recommend a graphing calculator (Ti-84 or Ti-Nspire); otherwise any scientific calculator (Ti-34 Multiview, Ti-36 Pro, or Ti-30Xs Multiview) works. Available for purchase at the Clark Bookstore or rent at the Library. ALEKS — required online homework platform, includes access to the Smith textbook. Paid through your course registration.
Companion Web Site	ALEKS (aleks.com), accessed through Canvas
Software	Microsoft Office (primarily Word and Excel) — accessible right inside Canvas through the Microsoft Education integration, using your free Clark student Office 365 account.
Electronic Resources	Canvas Learning Management System. You'll need to scan and upload multi-page PDF files to Canvas — Microsoft OneDrive or Adobe Scan are recommended. Follow the directions in the Getting Started: Course Orientation guide, which include enabling your Clark student email and Clark Labs account to activate your free Microsoft Office 365 account.
Privacy & accessibility for course tools and materials	See Clark's eLearning accessibility and privacy policies page for Canvas, ALEKS, and other tools used in this course.
Where to find necessary materials	Chemfolio and your lab manual: Clark College Bookstore. ALEKS: access is set up automatically in Canvas once your registration is processed.

What Are the Course Agreements?

TL;DR: Message me in Canvas Inbox, cite anything that isn't yours, never let AI write your discussion posts, and show up ready to work.

Communication

Canvas Inbox is our channel — for everything. I don't use email during the term, so if you send a Clark email, expect it to go unanswered; message me through Canvas Inbox instead. I respond within 24 hours on weekdays.

Citing Your Sources

It is expected that you are submitting your own work regardless of format. Citations in this class can be in APA 7, ACS, CSE, or if you must MLA Style. Written assignments in this course serve as tools to facilitate your growth as well as assessments to demonstrate achievement of course competencies. Your ideas and knowledge are important and cannot be replicated by others or by artificial

intelligence (AI). There are resources available to support you including your professors, Clark College Librarians, and writing tutors. They want to help!

Academic dishonesty often happens when students are stressed and feeling under pressure or when cultural differences lead to miscommunication around local customs. There are two main types of academic dishonesty. Plagiarism is when a student doesn't credit or properly cite the sources for their academic work. Cheating refers to a student getting or giving help on assignments that isn't allowed by the professor. Use of AI tools submitted as your own work will be considered both plagiarism and cheating.

A citation is when you give credit to another source of information that you are using in your work. This course has specific expectations about how you cite sources when you are referring to work that is not your own (for example, if you use an outside quote in a discussion post, etc.). To avoid plagiarism (when you intentionally or unintentionally present someone else's work as your own), you will need to signal that you are referencing outside material and give credit. Read the [plagiarism information from Clark College Libraries](#) to learn about what plagiarism is and how to avoid it. It has a tutorial, tips, and style guides for citations. Ask us for help on this - it's freely given!

AI (Artificial Intelligence) Guidelines — Community AI Agreement

TL;DR: No AI-generated content anywhere, in anything you submit — discussions, labs, ALEKS, quizzes, the Final Project. No exceptions, no declaring your way around it. Any AI use in a submission earns 20% on that assignment — a failing score — with repeated or more serious violations dropping to zero.

Writing is integral to thinking. It is also hard. In this class, I look forward to and am excited to see your original work and perspectives written in your own voice. You are not graded on grammar. Natural language processing (NLP) applications like ChatGPT, OpenAI, xAI, LLMs like DeepSeek, should be used with caution and only if you are documenting your work in Microsoft Word or Google Docs. You should note that the material generated by these programs may be inaccurate, incomplete, biased, or otherwise really crappy. Beware that use WILL stifle your own independent thinking and creativity. You cannot submit any work or even one sentence generated by an AI tool as your own. Even "brainstormed" material or ideas generated by AI should be credited instead of you, just like any other reference material. **Assignments with AI-generated or AI-edited content earn 20% — a failing score. Repeated or more serious violations may result in a zero on that assignment.**

Think of the help you get from NLP apps as a much less sophisticated version of the assistance you can receive (for free!) from the Clark College Librarians or Writing Center tutors. A librarian or tutor might legitimately ask you a question to jump-start your imagination, or identify a poorly organized paragraph, but should never do the writing for you. A major difference here is that an NLP app is not a person. It's a machine which is adept at recognizing patterns and reflecting those patterns back at us. It cannot think for itself. It cannot think for you. It is getting worse at writing for you.

We need honest folks in healthcare occupations. Here's how to avoid cheating:

- **Document ALL your work offline in Microsoft Word in your OneDrive (see Canvas).**
- **Appropriate use of AI for any assignment in this class:**
 - You are free to use spell check, grammar check, and synonym identification tools in applications (e.g., those in Microsoft Word, Google Docs, Pages, or Canvas).
 - You are free to use in-app recommendations when it comes to spelling and grammar of only sentences you have written entirely yourself first.
 - You are free to use in-app recommendations when it comes to tweaking outlines you have written entirely yourself first.
- **Inappropriate use of AI for any assignment in this class:**
 - You cannot have an app write a draft (either rough or final) or brainstorm any assignment.
 - You cannot use any content generated from prompts given to you in assignments.
 - You cannot use entire sentences or paragraphs suggested by an app without providing quotation marks and a citation, just as you would any other source (see Citing Your Sources above).

- You cannot have an app reply to your classmates or provide their words to an AI tool. **This is a violation of your classmates' privacy as a student at Clark.**
- You cannot give any AI tool the learning materials and assignments from this class. **This is a violation of my privacy, as these materials were made by us.**
- **Evidence of inappropriate AI use will be grounds for submission of an Academic Dishonesty report to the College, along with course sanctions at my discretion.**

Dr. B's Note: *You're entering health careers where accuracy and integrity matter. This policy isn't about distrust — it's about protecting your own growth as a thinker, which is the actual skill you're here to build.*

Classroom Norms

As your professor, I want to make clear my commitment to creating a welcoming and compassionate classroom culture in-person and online that supports a variety of thoughts, perspectives, and lived experiences. Healthcare occupations are founded on ethical and professional standards that are improved through the inclusion of diverse voices and perspectives. We value an inclusive environment, and harassment, bias, hate, and discrimination directed at a peer or professor will not be tolerated.

To honor these commitments, I expect that everyone — including me — will:

- Be Kind, Patient, Understanding
- Create a brave learning space
- Learning can be tough, discomfort is welcome
- Practice graceful learning
- Respect each other's perspectives
- Make space, take space
- Include others in the conversation
- Stay present, but take breaks when you need them
- Listen to understand, not speak next
- Stories told here stay here, knowledge leaves
- Accept reasonable boundaries and non-closure
- If someone is struggling, offer to help
- If you are struggling, seek help

Alongside that spirit, a few practical expectations:

- Complete each week's assignments — discussions, labs, ALEKS — by the posted due date.
- Attend your scheduled labs (both) every week; all lab hours are required to pass the course — see the Lab Requirement section below for exactly what that means.
- Be a courteous member of the team and don't get in the way of anyone else's learning: post and reply before the due dates, not hours or minutes before.
- Tell me as early as you can about anything affecting your ability to take part fully.

"How you do anything is how you do everything."

What Are the Learning Goals in This Course?

TL;DR: By the end, you can apply chemistry reasoning to real health-career problems — and these skills tie directly to Clark's General Education Competencies.

General Education at Clark College empowers students to become informed, adaptable, and engaged community members who are prepared to contribute to a more just and equitable society. Through the competencies below, students in this class build a foundation of skills and knowledge to achieve their personal and professional goals:

- **Critical Thinking 3** — Formulate well-reasoned arguments supported by evidence
- **Critical Thinking 4** — Apply creative thinking skills to generate new ideas and solve problems
- **Problem-Solving 1** — Accurately define a problem
- **Communication 3** — Apply communication skills and concepts to real-world situations
- **Civic & Global Consciousness 1** — Explain global issues, considering various perspectives
- **Civic & Global Consciousness 3** — Identify the requirements of responsible civic engagement

Student learning outcomes are the skills and abilities that you are working to acquire, practice, and apply in class. They represent the learning goals for the term.

Upon successful completion of this course, students will be able to:

- **CLO1:** Applies the fundamental rules behind atomic structure and element reactivity to predict how bonds are formed and break, with context to physiology, pharmacology, and metabolism.
- **CLO2:** Contrast the forces at play within and between molecules or ions with their emergent properties and water (hydrophilic) or fat solubility (hydrophobic) in medical or environmental contexts.
- **CLO3:** Performs scaled-thinking and mathematical calculations to simplify and evaluate contextual problems that impact the community or environment using quantitative information.
- **CLO4:** Compose opinions, interpretations, and solid evidence regarding chemical systems and determine their impact on society in everyday life.

The table below lists the learning outcomes for this course with the corresponding assessments, to support Clark's General Education Competencies:

Course Outcomes	Assessment of Course Outcomes
CLO1	ALEKS Problem Sets, 3-2-1 Team Posts, Lab Experiments, OOL Discussions, Final TYC Project
CLO2	Quizzes, ALEKS Problem Sets, 3-2-1 Team Posts, Lab Experiments, Final TYC Project
CLO3	Quizzes, ALEKS Problem Sets, 3-2-1 Team Posts, Lab Experiments, Final TYC Project
CLO4	OOL Discussions, Lab Experiments, Final TYC Project

How Will I Be Graded, and Why?

TL;DR: 3-2-1 Team Posts 15% · OOL Discussions 15% · Lab Experiments 25% · Quizzes 20% · ALEKS Problem Sets 15% · Final TYC Project 10%. Here's why those weights, and the full breakdown.

These weights aren't arbitrary. Lab Experiments carry the most weight because that's where the real demonstration of your learning happens — applying what you know, not just recognizing it. Quizzes are the next-largest individual measure of your understanding. 3-2-1 Team Posts and OOL Discussions are lighter individually, but combined (30%) they carry as much weight as Labs. ALEKS Problem Sets support your learning but aren't proof of understanding on their own, which is why they're weighted below Labs and

Quizzes. The Final TYC Project replaces a comprehensive exam because teaching a concept to someone else is a better test of whether you actually understand it than a timed exam is (e.g. “Learn one, do one, teach one.”)

Assignment Categories and Their Percentage of the Final Grade

Category	Weight	What it is
3-2-1 Team Posts	15%	Weekly 3-2-1 Team Discussions — lecture Activities completed and uploaded, plus your engagement with your team
OOL Discussions	15%	Weekly Outside-of-Lecture discussion post and replies
Lab Experiments	25%	Weekly on-campus labs — Background, Methods, Results, Findings, and Conclusions (100 points per lab)
Quizzes	20%	In-person, given during lab (Weeks 2, 5, 7, 9, and 10)
ALEKS Problem Sets	15%	Online adaptive homework for content practice
Final TYC Project	10%	Teach Your Community project — no final exam – details in the TYC Canvas module

You must earn at least 65% in every one of the six grading categories above to pass this course — not just an overall passing average. Falling below 65% in any single category, even with a passing overall grade, means you cannot pass.

About the Lab Requirement

Lab hours are part of this course's official credit hours — 1 of your 5 credits is earned specifically through lab. Because of that, each lab must genuinely demonstrate your work, not just check a box.

- Each lab has five parts, totaling 100 points: Background (10 pts) and Methods (40 pts), Results (20 pts), and Findings (10 pts) and Conclusions (20 pts) — see Weekly Flow above for exactly when each is due.
- Makeup labs are typically not possible. If you miss one experiment, notify me beforehand to be eligible for make-up data — you'll take an 11-point deduction for not collecting the data yourself, but can still complete the rest of the lab assignments for up to 89 of the 100 possible points.
- If you miss two experiments, the second missed lab is scored zero.
- Missing three weeks of lab, or having two incomplete labs, means you cannot pass the course.

Grading Scale

Note: Clark does not award C-, D+, or D- in this course — a C- does not transfer, so grading stops at a clean C, then D, then F.

Percent	Grade	Percent	Grade	Percent	Grade
93 up to 100	A	83 up to 87	B	73 up to 76	C
90 up to 93	A-	80 up to 83	B-	60 up to 73	D
87 up to 90	B+	77 up to 79	C+	0 up to 60	F

Late Work

Life happens during an academic term. I trust you know your situation and I don't need an explanation. My position is to support your learning and your success.

- All assignments remain open after the due date, losing 1% per hour until worth zero points (approximately 5 days). Assignments can still be submitted after this, but the score will be zero.
- In-person lab attendance and the Final TYC Project have no grace period.
- No due dates extend beyond the course end date.
- If something requiring a prolonged absence occurs, contact me via Canvas Inbox and let's see what solutions are possible.

Feedback

Discussions and other work are graded within 48 hours of each due date.

Revision

There is no separate revision or regrade policy at this time — the late-work window above functions as the built-in opportunity to catch up before work is scored.

What Resources and Support Services Are Available to Me?

TL;DR: Tech Hub for Canvas/software problems, Tutoring or Librarians for coursework help, Basic Needs Hub if life is making this term hard.

Technical Support

Item	Details
Tech Hub	Canvas, software, or account trouble — Scarpelli Hall 135 · techhub@clark.edu · 360-992-2010
To report a Canvas problem	The Help button in Canvas → "Report a Problem" (replies come to your Clark Gmail) · Canvas Guides · Student Guide
Setup help	Getting Started: Course Orientation has the Office/OneDrive setup guide — this step trips people up, so budget real time and reach out to Tech Hub if you hit an error

Academic Support

Item	Details
Student Success Coaches	One-on-one coaching for study/organization skills, emotional and personal support, and connections to college resources. PUB 002 or Zoom.
Clark College Tutoring Services	In-person and virtual tutoring via Zoom; see the Virtual Tutoring Center Canvas course; eTutoringOnline.org for additional online writing lab support.
Clark College Librarians	One-on-one help with research projects and source citations, in person or by chat .

Student Support

Item	Details
Counseling and Health Center	Physical and mental health services to support student wellness and academic success.

Item	Details
Office of People and Culture (OPC)	Supports systemically non-dominant groups navigating Clark College; the Diversity Center is a great place to visit between classes.
Penguin Pantry	Monthly pre-packaged food boxes. Contact Cath Busha, Dean of Student Engagement (cbusha@clark.edu).
Basic Needs Hub	Help with food, housing, and financial challenges — intake form . basicneeds@clark.edu · 360-992-2766 · Gaiser Hall (GHL) 216.
Advising Services	Co-create academic plans aligned with your career and personal goals — schedule an appointment .
Career Services	Resources and strategies for career and employment goals. 360-992-2902 · careerservices@clark.edu
Veterans Center of Excellence	Resources, community, and support for U.S. military service members. PUB 015 · 360-992-2073 · Vetresources@clark.edu

You can find all these resources as well as links to the [Clark bookstore](#), [C-Tran](#), [campus maps](#), and campus life and activities at the [Clark Students website](#).

How Should I Get Started in Class?

See Your First Week, near the top of this syllabus, for the exact five things to do before anything else.

Appendix: Official College Policies

TL;DR: The full text of Clark's required policy language lives in one centrally-maintained Canvas course, so it can't drift out of date here. Quick summaries below; contact me if any of these apply to you.

- **ADA Accommodations** — If a disability or temporary health condition affects your coursework, the [Disability Access Center](#) can set up accommodations — contact them as early as you can, then let me know within 24 hours so we can plan for this course.
- **Non-discrimination Policy** — Clark prohibits discrimination and harassment against any member of the college community; there's a formal grievance process through Human Resources if you experience it.
- **Title IX and Sexual Misconduct** — If you or someone you know has experienced sexual misconduct, you're not alone — support, reporting options, and campus resources are available.
- **Student Rights and Responsibilities** — Covers FERPA, absences for reasons of faith or conscience, and your general rights as a student.

Full policy text for all of the above: [Clark College Policies Canvas course](#).